

ANDRÉS JOSUÉ CASTRO ANGULO

Senior Software Engineer | Modern C++ & Python Specialist | Autonomous Systems & Backend

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PROFESSIONAL SUMMARY

Senior Software Engineer with a Master's degree in Mechatronics Engineering and over 9 years of experience designing, building, and deploying high-performance backend software, distributed systems, and autonomous robotics solutions. Proven expertise in Modern C++ (C++17/20), Python, ROS/ROS2 architecture, multithreaded low-latency optimization, and clean SOLID design across production environments.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Systems & Programming Languages:** Modern C++ (C++17), Python, C, JavaScript, Node.js, Multithreading, Concurrent Programming
- **Robotics & Autonomy:** ROS, ROS2, Kinematics & Dynamics, Autonomous Navigation, Motion Planning, Control Algorithms, SLAM
- **Perception & Compute:** OpenCV (Image Processing), CUDA, OpenCL, Sensor Integration, Multi-sensor Fusion
- **Software Architecture & Dev Tools:** Distributed Systems, REST APIs, SOLID Principles, Linux (Ubuntu, Debian, CentOS), Git, CI/CD, Docker, AWS, Azure
- **Databases:** PostgreSQL, MySQL, MariaDB, MongoDB, Redis

PROFESSIONAL EXPERIENCE

Sr. Software Engineer | Thomson Reuters

(July 2022 – July 2026)

- Engineered, optimized, and maintained mission-critical backend services and distributed systems using Modern C++ (C++17/20) in Linux environments.
- Optimized application latency, memory footprint, and concurrency execution for high-throughput production software.
- Performed deep-dive debugging, system profiling, and root-cause analysis to ensure maximum uptime, fault tolerance, and software stability.
- Enforced clean architecture, SOLID design principles, and comprehensive code reviews across cross-functional engineering teams.

Lead Mechatronics Engineer | Octopy

(May 2020 – June 2022)

- Led a multidisciplinary engineering team through the full lifecycle development and deployment of autonomous service robots.
- Architected ROS/ROS2 software nodes, control loops, and services in C++ and Python for state estimation, SLAM, and trajectory planning.
- Integrated multi-sensor arrays (cameras, LiDAR, encoders) and implemented computer vision pipelines using OpenCV for perception and safety.

Software Engineer / Sr. Software Engineer | SpringLabs

(March 2017 – May 2020)

- Led cross-functional engineering teams in building multiplatform software applications, REST APIs, and database integrations (PostgreSQL, MySQL).
- Enforced modular architecture design, peer code reviews, and technical mentorship across software projects.

EDUCATION

Master of Engineering in Mechatronics – Instituto Tecnológico de Mérida, Mérida, Yucatán

December 2019

Bachelor of Engineering in Mechatronics – Universidad Autónoma de Yucatán, Mérida, Yucatán

March 2014